

Multiplication and Inverse Matrices

Five ways to see a matrix product, all giving the same numbers. Then the inverse: when it exists, and how Gauss–Jordan finds it.

PART 1 Index

What this lesson covers, in order. Click a topic to jump to it.

1. The entry rule and shapes

Each entry of AB is a row of A times a column of B . Which shapes can be multiplied.

2. By columns and by rows

Columns of AB mix the columns of A ; rows of AB mix the rows of B . Many portfolios at once.

3. Columns times rows

AB as a sum of simple pieces, one per fund.

4. Block multiplication

Cut matrices into blocks and multiply the blocks.

5. The inverse, and when it exists

$A^{-1}A = I$. No inverse when some trade changes nothing: $Ax = 0$ with $x \neq 0$.

6. Gauss–Jordan: 2×2

Find A^{-1} by eliminating $[A \mid I]$ until it becomes $[I \mid A^{-1}]$.

7. Gauss–Jordan: 3×3 , and why it works

The three-fund portfolio, step by step. $E[A \mid I] = [I \mid E]$, so $E = A^{-1}$.

8. Using the inverse

Every target at once, building-block trades, the 2×2 formula, and when not to compute A^{-1} .

METHODS IN THIS LESSON

[Entry rule](#)
[Shapes rule](#)
[Column way](#)
[Row way](#)
[Column \$\times\$ row \(outer products\)](#)
[Block multiplication](#)
[Singular test \(\$Ax = 0\$ \)](#)
[Gauss–Jordan elimination](#)
[2 \$\times\$ 2 inverse formula](#)

PART 2 Prerequisites

Click any link to review it, then come back.

- Matrix $\times$ vector, two ways (Lesson 1): Ax mixes the columns of A .
- Singular matrices (Lesson 1): funds that add no new direction.
- Elimination on the augmented matrix (Lesson 2): the portfolio eliminated to U .
- Elimination matrices and "row $\times$ matrix mixes rows" (Lesson 2).
- Undoing one step (Lesson 2): the inverse of a single elimination matrix.

PART 3 Explanation

THE RUNNING EXAMPLE: MANY CLIENTS, ONE SET OF FUNDS

Same funds as always: Nifty (12%, beta 1.0), bonds (7%, beta 0), midcap (15%, beta 1.2). Up to now we multiplied the fund matrix A by **one** portfolio at a time. A desk handles many at once, so we stack the portfolios side by side as the columns of a matrix X :

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix}}_{A: \text{Nifty, bonds}} \quad \underbrace{\begin{bmatrix} 7 & 4 \\ 3 & 6 \end{bmatrix}}_{X: \text{client 1, client 2}} \quad (\text{₹ lakh in each fund})$$

Client 1 holds ₹7 lakh Nifty and ₹3 lakh bonds; client 2 holds ₹4 lakh and ₹6 lakh. The product AX gives every client's (budget, return) at once. Then we turn it around: given the targets, the **inverse** A^{-1} gives the portfolios.

1. The entry rule and shapes

METHOD · ENTRY RULE (ROW TIMES COLUMN)

The entry in row i , column j of $C = AB$ is the **dot product** of row i of A with column j of B : $c_{ij} = \sum_k a_{ik} b_{kj}$.

$$\begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix} \begin{bmatrix} 7 & 4 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} 10 & 10 \\ 105 & 90 \end{bmatrix}$$

Take c_{22} , row 2, column 2: (row 2 of A) $\cdot$ (column 2 of X) = $12 \times 4 + 7 \times 6 = 48 + 42 = 90$. That is client 2's return: ₹4 lakh in Nifty earns ₹48 thousand, ₹6 lakh in bonds earns ₹42 thousand.

RULE · SHAPES

An $m \times n$ matrix times an $n \times p$ matrix gives an $m \times p$ matrix. The inner sizes must match: **columns of A = rows of B** (one entry of each portfolio per fund). Here $(2 \times 2)(2 \times 2) = 2 \times 2$; three funds and five clients would be $(3 \times 3)(3 \times 5) = 3 \times 5$.

Strang's point for this lecture: there are several ways to see the same product, they all give the same numbers, and each one is useful.

2. By columns and by rows

METHOD · COLUMN WAY

Column j of AB is A times column j of B . So the columns of AB are combinations of the columns of A .

Each client's portfolio is one column of X , and A turns it into that client's (budget, return):

$$A \begin{bmatrix} 7 \\ 3 \end{bmatrix} = 7 \begin{bmatrix} 1 \\ 12 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ 7 \end{bmatrix} = \begin{bmatrix} 10 \\ 105 \end{bmatrix}, \quad A \begin{bmatrix} 4 \\ 6 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 12 \end{bmatrix} + 6 \begin{bmatrix} 1 \\ 7 \end{bmatrix} = \begin{bmatrix} 10 \\ 90 \end{bmatrix}.$$

This is Lesson 1's column picture, done for many clients side by side.

METHOD · ROW WAY

Row i of AB is row i of A times B . So the rows of AB are combinations of the rows of B (Lesson 2's "row $\times$ matrix mixes rows").

Rows of X are "₹ in Nifty for each client" and "₹ in bonds for each client":

row 1: $1(7, 4) + 1(3, 6) = (10, 10)$ (every client's budget), row 2: $12(7, 4) + 7(3, 6)$

Columns answer "what does each client get?"; rows answer "what is each rule across all clients?".

▼ Check yourself: a third client holds ₹5 lakh in each fund. What new column appears in AX ?

$5(1, 12) + 5(1, 7) = (10, 95)$: ₹10 lakh used, ₹95 thousand earned.

3. Columns times rows

A **column times a row** is not a number but a whole matrix:

$$\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 12 \\ 3 & 18 \\ 4 & 24 \end{bmatrix} \quad (\text{Strang's example})$$

Every column is a multiple of $(2, 3, 4)$ and every row is a multiple of $(1, 6)$. Strang calls this a "really minimal matrix": all its rows lie on one line, and so do all its columns.

METHOD · COLUMN-TIMES-ROW (SUM OF OUTER PRODUCTS)

$AB = (\text{column 1 of } A)(\text{row 1 of } B) + (\text{column 2 of } A)(\text{row 2 of } B) + \dots$. Each term, a column times a row, is called an **outer product**.

For the portfolios, this splits AX by fund:

$$\underbrace{\begin{bmatrix} 1 \\ 12 \end{bmatrix} \begin{bmatrix} 7 & 4 \end{bmatrix}}_{\text{Nifty's part}} + \underbrace{\begin{bmatrix} 1 \\ 7 \end{bmatrix} \begin{bmatrix} 3 & 6 \end{bmatrix}}_{\text{bonds' part}} = \begin{bmatrix} 7 & 4 \\ 84 & 48 \end{bmatrix} + \begin{bmatrix} 3 & 6 \\ 21 & 42 \end{bmatrix} = \begin{bmatrix} 10 & 10 \\ 105 & 90 \end{bmatrix}.$$

The first piece is "what the Nifty holdings contribute to every client's budget and return", the second is the bonds' contribution. This kind of **attribution by source** is everyday work in portfolio analysis, and it is just the column-times-row view of a matrix product.

4. Block multiplication

METHOD · BLOCK MULTIPLICATION

Cut A and B into blocks whose sizes match, then multiply the blocks as if they were numbers: $\begin{bmatrix} A_1 & A_2 \\ A_3 & A_4 \end{bmatrix} \begin{bmatrix} B_1 & B_2 \\ B_3 & B_4 \end{bmatrix}$ has top-left block $A_1B_1 + A_2B_3$.

With three funds and three clients, cut the fund matrix into "Nifty and bonds" | "midcap", and the client matrix the same way by rows:

$$\underbrace{\left[\begin{array}{cc|c} 1 & 1 & 1 \\ 12 & 7 & 15 \\ 1 & 0 & 1.2 \end{array} \right]}_{\begin{bmatrix} A_{\text{left}} & A_{\text{right}} \end{bmatrix}} \underbrace{\left[\begin{array}{ccc} 5 & 4 & 6 \\ 3 & 5 & 6 \\ 2 & 1 & 3 \end{array} \right]}_{\begin{bmatrix} X_{\text{top}} \\ X_{\text{bottom}} \end{bmatrix}} = A_{\text{left}} X_{\text{top}} + A_{\text{right}} X_{\text{bottom}}$$

That is "the Nifty-and-bond part of every client" plus "the midcap part of every client". The total is

$$AX = \begin{bmatrix} 10 & 10 & 15 \\ 111 & 98 & 159 \\ 7.4 & 5.2 & 9.6 \end{bmatrix},$$

the three client targets from Lesson 4, each column one client's (budget, return, exposure). Large systems are multiplied in blocks like this so they fit in a computer's fast memory.

▼ **Check yourself (MIT problem 3.1):** with $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $C = \begin{bmatrix} 0 & 0 \\ 5 & 6 \end{bmatrix}$, compare $AB + AC$ with $A(B + C)$.

$AB = \begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix}$, $AC = \begin{bmatrix} 10 & 12 \\ 20 & 24 \end{bmatrix}$, sum $\begin{bmatrix} 11 & 12 \\ 23 & 24 \end{bmatrix}$. And $A(B + C) = A \begin{bmatrix} 1 & 0 \\ 5 & 6 \end{bmatrix} = \begin{bmatrix} 11 & 12 \\ 23 & 24 \end{bmatrix}$. Equal: matrix multiplication is **distributive**.

5. The inverse, and when it exists

Multiplication turns portfolios into targets. The **inverse** goes the other way.

DEFINITION: INVERSE

A square matrix A is **invertible** (non-singular) if there is a matrix A^{-1} with $A^{-1}A = I = AA^{-1}$. For square matrices a left inverse is automatically a right inverse.

Strang: whether A has an inverse is "the most important question you can ask about the matrix". Start with a matrix that doesn't.

When there is no inverse

The two-Nifty-funds matrix from Lesson 1, $A = \begin{bmatrix} 1 & 1 \\ 12 & 12 \end{bmatrix}$. Strang's favourite reason it can't be inverted:

$$\begin{bmatrix} 1 & 1 \\ 12 & 12 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The trade "₹1 lakh more in fund A, ₹1 lakh less in fund B" changes nothing: same budget, same return. Now suppose A^{-1} existed. Multiply both sides by it: $A^{-1}Ax = A^{-1}0$, so $x = 0$. But $x = (1, -1)$ is not zero. Contradiction: no inverse.

RULE · SINGULAR TEST: $AX = 0$ WITH $X \neq 0$

If some **non-zero** x has $Ax = 0$, then A has no inverse. In portfolio terms: if some trade changes nothing, the answer to "which portfolio hits this target?" can never be unique, so it can't be undone. Quick signs (from the recitation): a row or column of zeros, or two equal rows or columns.

Same with three funds: the **balanced fund** (half Nifty, half bonds) gives $A(1, 1, -2) = 0$. Buy ₹1 lakh each of Nifty and bonds and sell ₹2 lakh of the balanced fund, and nothing changes.

There's also a column reason: every column of AX is a combination of A 's columns, which all lie on one line. The columns of I , $(1, 0)$ and $(0, 1)$, don't both lie on that line, so $AX = I$ is impossible.

6. Gauss–Jordan: 2×2

Now a matrix that **does** have an inverse, the two-fund matrix. Write A^{-1} 's columns as unknowns:

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix}}_A \underbrace{\begin{bmatrix} a & c \\ b & d \end{bmatrix}}_{A^{-1}} = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_I$$

By columns: $A(a, b) = (1, 0)$ and $A(c, d) = (0, 1)$. Finding the inverse is **solving two systems with the same A** . Jordan's idea: solve them together by augmenting with the whole identity.

METHOD · GAUSS–JORDAN ELIMINATION

Start from $[A \mid I]$. **Gauss**: eliminate downward until the left half is upper triangular. **Jordan**: keep going upward to clear above each pivot. Finally divide each row by its pivot. When the left half is I , the right half is A^{-1} . Named after Carl Friedrich Gauss and Wilhelm Jordan, a 19th-century German geodesist who popularised the upward step.

$$\left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 12 & 7 & 0 & 1 \end{array} \right] \xrightarrow{R_2 - 12R_1} \left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & -5 & -12 & 1 \end{array} \right]$$

Gauss is done (upper triangular). **Jordan**: clear the 1 above the second pivot with $R_1 - (1/-5)R_2 = R_1 + 0.2R_2$:

$$\left[\begin{array}{cc|cc} 1 & 0 & -1.4 & 0.2 \\ 0 & -5 & -12 & 1 \end{array} \right] \xrightarrow{R_2 \div (-5)} \left[\begin{array}{cc|cc} 1 & 0 & -1.4 & 0.2 \\ 0 & 1 & 2.4 & -0.2 \end{array} \right]$$

$$A^{-1} = \begin{bmatrix} -1.4 & 0.2 \\ 2.4 & -0.2 \end{bmatrix}.$$

Check $A^{-1}A$: row 1 is $-1.4(1, 1) + 0.2(12, 7) = (-1.4 + 2.4, -1.4 + 1.4) = (1, 0) \checkmark$; row 2 is $2.4(1, 1) - 0.2(12, 7) = (0, 1) \checkmark$.

▼ Check yourself (Strang's example): invert $\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$ by Gauss–Jordan.

$R_2 - 2R_1$ gives $\begin{bmatrix} 1 & 3 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{bmatrix}$; $R_1 - 3R_2$ gives $\begin{bmatrix} 1 & 0 & 7 & -3 \\ 0 & 1 & -2 & 1 \end{bmatrix}$. So $A^{-1} = \begin{bmatrix} 7 & -3 \\ -2 & 1 \end{bmatrix}$.

7. Gauss–Jordan: 3×3 , and why it works

The three-fund portfolio. Start from $[A \mid I]$:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 12 & 7 & 15 & 0 & 1 & 0 \\ 1 & 0 & 1.2 & 0 & 0 & 1 \end{array} \right]$$

Gauss: downward (the same steps as Lesson 2)

$R_2 - 12R_1$, then $R_3 - R_1$, then $R_3 - 0.2R_2$:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -5 & 3 & -12 & 1 & 0 \\ 0 & 0 & -0.4 & 1.4 & -0.2 & 1 \end{array} \right]$$

Look at the right half: it is exactly the elimination matrix $E = E_{32}E_{31}E_{21}$ from Lesson 2, messy 1.4 and all. The identity has quietly recorded every step.

Jordan: upward

- Clear above the third pivot -0.4 . Row 2: $R_2 - \frac{3}{-0.4}R_3 = R_2 + 7.5R_3$ gives $(0, -5, 0 \mid -1.5, -0.5, 7.5)$.
- Row 1: $R_1 - \frac{1}{-0.4}R_3 = R_1 + 2.5R_3$ gives $(1, 1, 0 \mid 4.5, -0.5, 2.5)$.
- Clear above the second pivot -5 . Row 1: $R_1 + 0.2R_2$ gives $(1, 0, 0 \mid 4.2, -0.6, 4)$.

Divide by the pivots

Row 2 $\div (-5)$ and row 3 $\div (-0.4)$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 4.2 & -0.6 & 4 \\ 0 & 1 & 0 & 0.3 & 0.1 & -1.5 \\ 0 & 0 & 1 & -3.5 & 0.5 & -2.5 \end{array} \right] \Rightarrow A^{-1} = \begin{bmatrix} 4.2 & -0.6 & 4 \\ 0.3 & 0.1 & -1.5 \\ -3.5 & 0.5 & -2.5 \end{bmatrix}.$$

Check one column: $A \times (4.2, 0.3, -3.5)$: budget $4.2 + 0.3 - 3.5 = 1$; return $50.4 + 2.1 - 52.5 = 0$; exposure $4.2 + 0 - 4.2 = 0$. That is column 1 of I ✓.

WHY GAUSS-JORDAN WORKS

All the row operations together are one matrix E , applied to both halves: $E[A \mid I] = [EA \mid E]$. We stopped when the left half was I , so $EA = I$, which means $E = A^{-1}$. And the right half is $EI = E$: it has to be A^{-1} .

8. Using the inverse

Every client at once

If $AX = B$ (portfolios X hit targets B), then multiplying by A^{-1} gives $X = A^{-1}B$. For the three Lesson 4 clients:

$$A^{-1} \begin{bmatrix} 10 & 10 & 15 \\ 111 & 98 & 159 \\ 7.4 & 5.2 & 9.6 \end{bmatrix} = \begin{bmatrix} 5 & 4 & 6 \\ 3 & 5 & 6 \\ 2 & 1 & 3 \end{bmatrix} :$$

client 1 gets $(5, 3, 2)$, client 2 gets $(4, 5, 1)$, client 3 gets $(6, 6, 3)$, all in one product.

The columns of A^{-1} are building-block trades

Since $AA^{-1} = I$, column j of A^{-1} is the trade that changes rule j by exactly 1 and leaves the other rules unchanged:

Column of A^{-1}	Trade (Nifty, bonds, midcap, ₹ lakh)	Effect
1	$(4.2, 0.3, -3.5)$	+₹1 lakh budget, same ₹ return, same exposure
2	$(-0.6, 0.1, 0.5)$	+₹1 thousand return, same budget, same exposure
3	$(4, -1.5, -2.5)$	+1 exposure, same budget, same return

So a client who wants ₹5 thousand more return, nothing else changed should trade $5 \times (-0.6, 0.1, 0.5) = (-3, 0.5, 2.5)$: move ₹3 lakh out of Nifty, ₹0.5 lakh into bonds and ₹2.5 lakh into midcap. Desks call such columns **sensitivities**: how much each position must move per unit change in a target.

RULE · 2×2 INVERSE FORMULA

$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$, valid when $ad - bc \neq 0$. For the two funds: $ad - bc = 7 - 12 = -5$, giving $\frac{1}{-5} \begin{bmatrix} 7 & -1 \\ -12 & 1 \end{bmatrix}$, the same answer as Gauss-Jordan. The number $ad - bc$ is the *determinant* (much later in the course).

With it, Lesson 1's formula drops out: $x = -1.4B + 0.2R = (R - 7B)/5$ and $y = 2.4B - 0.2R = (12B - R)/5$ for any budget B and return R .

WATCH OUT: YOU RARELY NEED A^{-1} ITSELF

To solve $Ax = b$, don't compute A^{-1} and multiply. Factor $A = LU$ once and do two triangular solves (Lesson 4): it is cheaper and more accurate. Compute A^{-1} when you want its entries themselves, such as the sensitivities above.